

Activity-1

① Medical Image Enhancement.

Q1:- It would recommend Gamma.

(Power-Law) Transformation with $\gamma < 1$.

Q2:- Gamma transformation increases the brightness of dark regions without greatly affecting already bright regions. In an x-ray image, bones are already bright, but soft tissues appear dark. This transformation makes the soft tissues more visible, helping doctors detect abnormalities easily.

Q3:- The darker tissues become brighter and clearer. The overall image quality improves, making both bones and soft tissues easier to examine.

② Night-Time Surveillance.

Q1:- Log Transformation is the best choice.

Q2:- Since most pixel values are very low (0-40) the image is too dark. Log transformation expands these low-intensity values, making the

image brighter. negative transformation only reverse the brightness values and does not improve the visibility of dark areas.

Q3:- The enhanced image will be brighter with better visibility. objects such as car, people, roads, and parking lines will become clearer making surveillance more effective.

③ Fingerprint Recognition.

Q1:- Contrast stretching.

Q2:- Fingerprint images often have low contrast between ridges and valleys. Contrast stretching increases the difference b/w these gray levels, making the fingerprint pattern sharper and easier to recognize.

Q3:- Log Transformation would brighten the dark parts to the fingerprint, but it would not improve the ridge and valley contrast. as effectively as contrast stretching. As a result, some fingerprint details may become

less distinct, reducing recognition accuracy.